

# Two Class Motor Imagery Classification Based on ANFIS

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**Abstract.** Mental task or action performed in imagination is called Motor Imagery (MI). For rehabilitation therapy of movement disorders and improvement in motor skill MI is utilized as a promising cognitive tools. Many people suffering from amyotrophic lateral sclerosis (ALS), spinal cord injury and other neurological diseases in which, motor cells in the spinal cord, brainstem are progressively deteriorated and die. In this conditions patients are not able to perform any muscular activity. To solve this kind of problem, patient performs a motor imagery task with a brain-computer interface (BCI). It provides an interface for conversion of brain signals into communication and control signal by performing pre-processing and pattern recognition task. In BCI accurate classification of the brain signals with the least misclassification error is a crucial challenge. There are many techniques are explored in the area of motor imagery classification still it suffers from low accuracy and computational inefficiency. This paper present better pre-processing and classification for motor imagery based EEG signals. Discrete wavelet transform (DWT) features are computed from decomposed coefficients. Equal width discretization (EWD) and Shannon entropy were used to discretize and for selection of features respectively. After that selected features are applied as input into ANFIS for classification. We create two models based on sub-clustering and fuzzy c-means clustering and perform the comparative analysis. The results confirm that the ANFIS model based on sub-clustering gives better accuracy as compared to fuzzy c-means clustering method. Model achieved accuracy between 62-69% for all four subjects. The performance of ANFIS is better as compared to other work mention in literature.

**Keywords:** Adaptive Neuro-Fuzzy Inference System (ANFIS) · Electroencephalograph (EEG) · Discrete Wavelet Transform (DWT) · Equal Width Discretization (EWD) · Shannon entropy. sub clustering · Fuzzy c-means clustering

## 1 Introduction

Nowadays many people are suffering from the neurological disorder. These people are paralyzed due to high-level spinal cord injury so they are not able to perform any muscular activity. To solve this problem Some researchers have found EEG signal based communication system that is called Brain-computer interface (BCI). EEG

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signals are used as a control signal in BCI. Electroencephalograph(EEG) is a non-invasive method that used for the acquisition of brain signals [1]. The first time EEG is recorded in animal Brain in 1875 by Richard Coton. Han Berger (1924) [2] found that electrical signals produced by brain can be recorded from the scalp. Over a further figured conducted by, Han Berger and Herbert Jasper (1938) found that EEG could also be utilized for human communication [3]. Recording of EEG signal taken place using 10–20 EEG placement system on scalp using connection between location of electrode and latent area of brain and it has low resolution [4–6].

Brain computer interface (BCI) provides digital interface between the brain and the computer without any muscle activity [7, 8]. So it proves to be useful for communication of patients whose suffering from neurological deceases. Based on placement technique of electrode BCI is of two type: invasive and noninvasive. Each has its advantages and disadvantages given in [9, 10]. To constructing stable and reliable BCI require effective classification method. Perform classification of features value extracted from EEG signal leading to a lower accuracy. So in order to aspect a better classification accuracy we need a more precise and optimized operating interface. Many algorithms are used for classification of EEG signals but these classifiers are not able to handle non-stationary and non-linear nature of EEG signals. Fuzzy is a powerful tool to handle non-linearity of data. So here we have used fuzzy based ANFIS classifier for classification.

## 2 Literature Review

Wieyen Hsu [11] proposed an adaptive electroencephalograph analysis system to facilitate classification of two sessions, single-trial motor imagery data. Enhanced active segment selection further refining the active segment is reducing the event-related windows length to that of a 1-s segment. It also increases the accuracy. Finally, adaptive LDA with active enhanced segment selection increases the accuracy of classifier. Guo et al. [12] Applied genetic programming to extract the feature automatically with a goal of improving the classification accuracy of K-nearest neighbor (KNN) classifier and dimensionality reduction of the input feature. The proposed method also has some limitation: When dimension of original feature high then perform a GP based feature extraction is computationally more expensive. García-Laencina et al. [13] Found that adaptive design of BCI is important challenge in classification of motor imagery. For solving this problem Authors explores the dimension reduction techniques. They find that LFDA is perform better classification for varied subjects along with reduced computational complexity. Luo et al. [14] Proposed new dynamic frequency feature selection (DFFS). Firstly, the weights are set to the same settings. After the feature selection a decrease in the weight of the correctly classified samples was observed. Dalhoumi et al. [15] Used the frame work called online inter-subjects classification. The author uses a weighted average ensemble approach in order to classify the EEG signal. The limitation of the online Adoption method is limited to binary. Nayak et al. [16] perform classification of two class motor imagery on right hand and feet movement. authors used ‘HAAR’ wavelet feature extraction method for extracting detailed coefficients and an approximate coefficient

which reduces the dimension of the acquired signal that is then acquainted with a number of classifiers. Sahu et al. [17, 18] proposed two methodologies, in first method [19] they combine wavelet feature and statistical and perform the classification using different classifier. In second method [20], various other features extracted such as wavelet, short time Fourier transform, statistical are extracted by the application of various extraction methods.

In this work, we are going to use an adaptive neuro-fuzzy inference system for classification of EEG signals. ANFIS is the combination of Fuzzy logic and Neural network. Where fuzzy logic having the ability to take a decision but not able to perform learning. In case of a neural network, it able to learn but not able to take decision. So by combining both technologies make more powerful technology.

This paper organized as follows, in Sect. 2 Proposed work is described. In Sect. 3 discussed ANFIS and structure. In Sect. 4 the evaluation procedure and the experimental result and result are concluded in Sect. 5.

## 3 Proposed Work

In this section describe the proposed method, material and technique, which is used in this work. Complete detail are as follows:

### 3.1 Methodology

In this work we proposed method based on adaptive neuro-fuzzy inference system (ANFIS) for classification of two class motor imagery. In this method first, we perform some pre-processing method such as feature extraction through DWT, Discretization using equal width discretization (EWD) and feature selection using Shannon entropy. Then apply set of selected features in ANFIS models based on sub-clustering and fuzzy c-means clustering. Flow diagram of proposed method shown in Fig. 1.

### 3.2 Material and Method

In this sub-section describe the material and method used in our work. Complete detail of material and method are shown as below:

**Dataset**

Dataset taken from BNCI horizon 2020 [19]. Datasets belong to two class motor imagery. As per Graz BCI training paradigm a single session is carried out for recording, training and feedback. From the total 8 runs 5 are used for training and 3 are used for testing. Out of total 20 trails for a set, per class 50 trails are for training and 30 are for testing.

**Feature Extraction**

Feature extraction is the process of transforming large no of features into small set of features. For better representation of EEG signals there are various Feature extraction methods such as Fast Fourier Transform (FFT), Wavelet Transform, Principle

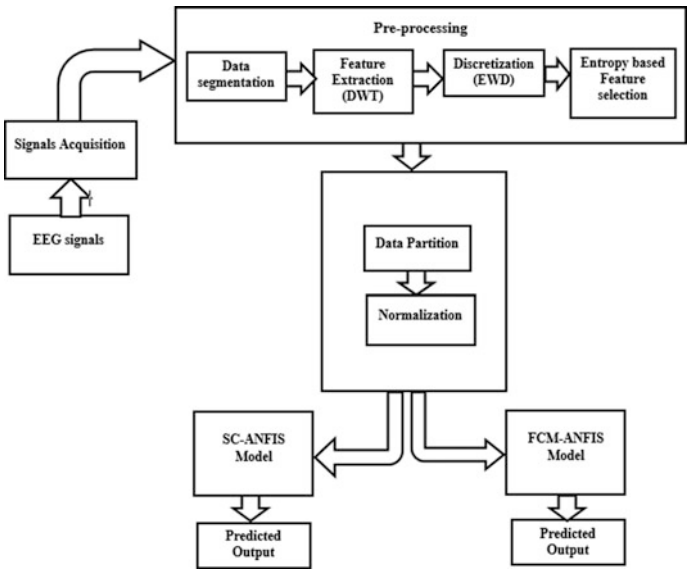



**Fig. 1.** Flowchart of proposed methodology

Component Analysis (PCA), Independent Component Analysis (ICA), Auto-Regression (AR) etc. Among all method WT widely used feature extraction method because its provide both time and frequency analysis. WT represent EEG signals more accurate. In our work we used DWT feature extraction method which described below in detail:

*Discrete Wavelet Transform*

DWT method is mostly used method which provide time-frequency analysis for better understanding of signals [20, 21]. It facilitates multi resolution analysis of EEG signals. The signals are split up into two part using high pass filter and low pass filter simultaneously. These filters are satisfying some condition called admissibility condition. DWT coefficient can be calculated using formula:

$$A_i = \frac{1}{\sqrt{M}} \sum_n x(n) \times \emptyset_{j,k}(\mathrm{n}) \tag{1}$$

$$D_i = \frac{1}{\sqrt{M}} \sum_n x(n) \times \psi_{j,k}(\mathrm{n}) \tag{2}$$

where $A_i$ and $D_i$ are approximate and detailed coefficient respectively, $\emptyset_{j,k}(\mathrm{n})$ and $\psi_{j,k}(\mathrm{n})$ represent dilation function and wavelet function. Here $M$ is a length of signal.

**Discretization**

Discretization is the process of partitioning attributes into set of features that generate different value for each attribute. This algorithm is used for faster and precise understanding of data [22]. There are two type discretization, unsupervised and supervised. Equal width (EWD) and Equal frequency (EFD) discretization is the easiest method for unsupervised discretization. In this paper, we are using EWD. Calculation of width size: The EWD method divides the values between maximum and minimum into K equal intervals. These intervals have the width of:

$$W = \frac{F_{MAX} - F_{MIN}}{K} \tag{3}$$

Here $W$ is width and $F_{MAX}$ is the maximum value of an attribute and $F_{MIN}$ is the minimum value of an attribute, $K$ is a number of the interval. The cut-points of the amplitude axis become:

$$F_{MIN} + W, F_{MIN} + 2W \ldots F_{MIN} + (K-1)W$$

**Feature Selection**

Feature selection is the process of selecting a relevant and informative feature from a set of feature. Select only those features that are useful to build a good predictor. In our case we are using Shannon entropy-based feature selection, however, entropy is a measure of uncertainty, that means a feature having high entropy consist high uncertainty so we only select those feature who have less entropy value.

## 4 Classification

Fuzzy logic and neural network are complementary to each other rather than competitive for system identification. Therefor it more beneficiary to combine these techniques. Here concept of hybrid system comes into scenario. ANFIS classifier: The ANFIS described by Jang [23]. Which is the combination of fuzzy logic and neural network. Where fuzzy logic performs reasoning process on the basis of fuzzy rules and membership function and neural network train all the parameter of antecedent and consequent by using learning algorithm.

The learning is performed by gradient descend or Hybrid method, ANFIS widely used in the biomedical field for analysis of the signal. It is the standard method for transforming human knowledge or experience into the rule base and database of the fuzzy inference system. ANFIS learn feature from training data and update its parameter according to given error criteria. FIS consists of five functional blocks (Table 1).

**Table 1.** Functions of FIS

| Functional block | Function |
|---|---|
| Rule base | Containing if-then rules |
| Database | Defining membership function |
| Decision making unit | Performing inference operation on the rule |
| Fuzzification interface | Converting crisp value into membership value |
| Defuzzification interface | Converting the fuzzy value into crisp value |

## 5 Results

In this section results obtain from preprocessing and classification method is discussed. In the proposed work first, EEG signals were divided into sub-band frequencies such as approximate and detailed coefficient by using DWT. An 18 features that are extracted using DWT, used as input for ANFIS classifier. Before classification preprocessing technique named discretization which is used for converting continuous value into discrete value. Further Shannon entropy is used for feature selection in which only the features having less entropy is selected.

### 5.1 Classification Results Using ANFIS

Selected features are used as input in ANFIS. Each feature having a 160 instances. Were these instances are divided into two data part: Training data and Testing data. Training data were used for training the ANFIS while testing data were used for evaluate the performance of ANFIS. Training data part containing 80% of data and remaining 20% were used as testing data. Inputs were Normalized using Min-Max function before it applying as input. First initial FIS are created on the basis of the rules which are extracted from two different algorithms: sub-clustering and fuzzy c-means clustering.

In sub clustering-based model Rules are generated by using Genfis2.m function in MATLAB®. First perform sub-clustering on training data and find the cluster centers.

Use these cluster centers in order to generate rules and determine input Gaussian membership function. Then use hybrid method to tune the rules. We are considering influence range 0.3 for finding the cluster. Giving the different inputs feature $F_i$ where i = 3, 4, 5, 6, observe performance of classifier for each feature. We found when number feature increases accuracy of classifier is also increase. Result are shown in the Table 2.

**Table 2.** Performances of ANFIS (sub clustering based)

| No. of features | S01 | | S02 | | S03 | | S04 | |
|---|---|---|---|---|---|---|---|---|
| | Training (%) | Testing (%) | Training (%) | Testing (%) | Training (%) | Testing (%) | Training (%) | Testing (%) |
| 4 | 65.28 | 43.75 | 70.14 | 43.75 | 56.00 | 50.00 | 56.94 | 37.50 |
| 5 | 70.14 | 43.75 | 81.25 | 50.00 | 59.72 | 56.25 | 70.14 | 62.50 |
| 6 | 80.56 | 62.50 | 88.19 | 62.50 | 59.03 | 56.25 | 80.56 | 62.50 |
| 7 | 90.28 | 68.75 | 95.14 | 68.75 | 64.58 | 62.50 | 90.28 | **62.50** |
| 8 | 93.06 | **68.75** | 97.22 | **68.75** | 74.51 | **68.75** | 93.06 | 56.25 |

In subject 1 (S01), subject 2 (s02), and subject 3 (S03) we achieve maximum accuracy 68.75% for 8 set of features, and in subject 4 (S04) we got maximum accuracy 62.50% for 7 set of features. Performance of ANFIS for different subject shown in Table 2. In the fuzzy c-means clustering based model we are using the Genfis3.m function in MATLAB® to generating a rule. Hybrid estimation is used for tuning the rule and membership function. we are generating different cluster and analysis the performance for each selected feature. The results are shown in the Table 3.

**Table 3.** Performance of ANFIS (Fuzzy c-means clustering)

| Subject | No. of features | No. of cluster | | | | |
|---|---|---|---|---|---|---|
| S01 | | 2 | 3 | 4 | 5 | 6 |
| | 4 | 37.50 | 56.25 | 50.00 | 37.50 | 37.00 |
| | 5 | 25.00 | 45.75 | 50.00 | 31.25 | 25.00 |
| | 6 | 50.00 | 43.75 | 37.50 | 43.75 | 37.50 |
| | 7 | 43.75 | 56.25 | 37.50 | 37.50 | 3750 |
| | 8 | 68.75 | 50.00 | 56.25 | 56.25 | 37.50 |
| S02 | 4 | 43.75 | 43.75 | 43.75 | 43.75 | 50.00 |
| | 5 | 43.75 | 43.75 | 56.25 | 25.00 | 31.25 |
| | 6 | 37.50 | 56.25 | 31.25 | 37.50 | 50.00 |
| | 7 | 68.75 | 56.25 | 68.75 | 50.00 | 43.75 |
| | 8 | 68.75 | 50.00 | 56.25 | 42.25 | 37.50 |
| S03 | 4 | 50.00 | 50.00 | 50.00 | 50.00 | 50.00 |
| | 5 | 56.25 | 56.25 | 56.25 | 56.25 | 56.25 |
| | 6 | 62.75 | 62.50 | 62.50 | 62.50 | 68.75 |
| | 7 | 50.00 | 62.50 | 62.50 | 37.50 | 62.50 |
| | 8 | 68.75 | 68.75 | 50.00 | 56.25 | 62.50 |
| S04 | 4 | 50.00 | 31.25 | 18.75 | 18.75 | 12.50 |
| | 5 | 50.00 | 62.50 | 37.50 | 50.00 | 68.75 |
| | 6 | 50.00 | 68.75 | 43.75 | 43.75 | 37.50 |
| | 7 | 31.25 | 50.00 | 50.00 | 50.00 | 37.50 |
| | 8 | 43.75 | 62.50 | 50.00 | 56.25 | 37.50 |

## 6 Conclusion

In this paper, we perform classification on a non-linear dataset of EEG signals by using ANFIS. we use two model based on sub-clustering and fuzzy c-means clustering. In case of the sub-clustering based model we find that when we increase the number of features then accuracy will also increase. we have performed classification on four subject S01, S02, S03 and S04 respectively. When we use the fuzzy c-means clustering based ANFIS, it gives random accuracy when increases the number of features and number of fuzzy logic, so we can say that for dataset sub-clustering based ANFIS is more consistent then fuzzy c-means clustering based ANFIS, and it gives a better result. Also compared with other work (Table 4) in literature we find proposed method perform better as compared to other proposed methods.

**Table 4.** Comparisons with other work

| Literature | S01 | S02 | S03 | S04 |
|---|---|---|---|---|
| Nayak et al. [16] | 45.3 | 71.8 | 56.2 | 46.8 |
| Sahu et al. [17] Based on PCA | 59.3 | 60.9 | 61.3 | Not performed |
| Sahu et al. [18] Based on Feature selection | 59.3 | 56.2 | 57.5 | 60.0 |
| Proposed model | 68.75 | 68.75 | 68.75 | 62.50 |

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